ECO1 – CONTINUUM COMPONENTS AND CLUSTERING

Abstract. This outline follows the proof of ECM1: a translation-invariant law of finite-component tempered fields in three dimensions, invariant and reflection positive at nine planes, clusters relative to its invariant sigma-algebra. Almost every ergodic component retains those reflections and clusters. A separate spatial-average argument transfers an additional Gaussian bound with its original constant. The linked map distinguishes the spectral, conditional-law, and Gaussian arguments.

Contents

1. Introduction and target statement 1
Proof overview 3
2. D. Field laws and the application hypotheses 4
3. M11. Reconstruction at one reflection plane 5
4. M12. The sharp coordinate cone from two mirrors 6
5. I. Invariant observables and zero energy 7
6. M13. Clustering uniformly in transverse displacement 8
7. E. Conditional laws and the common ergodic decomposition 9
8. M14. Reflection forms in the same ergodic components 10
9. G. The same Gaussian constant after conditioning 11

1. Introduction and target statement

We first state the continuum-law theorem, with its full hypotheses. The subsequent pages give the estimates and proof tasks in ECM1.

B2. Components and clustering

Interface. Overview Target.

Let V be a finite-dimensional real inner-product space with an orthogonal representation $\rho : O(3) \to O(V)$, and equip $\mathcal{X} = \mathcal{S}'(\mathbb{R}^3; V)$ with its evaluation sigma-algebra. For $g(x) = Qx + b$, set

$$(g \cdot f)(x) = \rho(Q) f(Q^{-1}(x - b)), \quad (g \cdot F)(f) = F(g^{-1} \cdot f), \quad \widehat{g}A(F) = A(g^{-1} \cdot F).$$

Write $U(a) = \widehat{\tau}_a$ and $\mathcal{R}_r = \widehat{r}$. For open O, let Σ_O be generated by $F(f)$ for real $f \in C_c^\infty(O; V)$. An r-invariant law is reflection positive at r, on the positive side H^+, when $\mathbb{E}[\overline{\mathcal{R}_r A} A] \geq 0$ for every finite character sum $A = \sum_b z_b e^{iF(f_b)}$, with real $f_b \in C_c^\infty(H^+; V)$.

Target theorem (ECM1, Theorem A). Let λ be a translation-invariant probability law on $\mathcal{X}$, invariant and reflection positive at the nine planes

$$x_j = 0 \quad (1 \leq j \leq 3), \qquad x_j - x_k = 0, \quad x_j + x_k = 0 \quad (1 \leq j < k \leq 3).$$

Choose the positive sides by making the displayed linear functions positive. Let $\mathcal{I}$ be the translation-invariant sigma-algebra, with invariance modulo λ-null sets, and put $P = \mathbb{E}_\lambda[\,\cdot \mid \mathcal{I}]$. Then, for all $A, B \in L^2(\lambda)$,

$$\langle A, U(a)B \rangle \longrightarrow \langle A, PB \rangle \qquad (|a| \longrightarrow \infty).$$

The distribution of the conditional law gives a unique decomposition $\lambda = \int \lambda_\xi \, d\nu(\xi)$ into translation-ergodic laws. On one common ν-full set, each component is invariant and reflection positive at every

real translate of the nine planes, is cubically invariant, and satisfies

$$\langle A, U(a)B\rangle_{L^2(\lambda_\xi)} \longrightarrow \overline{\mathbb{E}_{\lambda_\xi} A}\, \mathbb{E}_{\lambda_\xi} B \qquad (|a| \longrightarrow \infty)$$

for every $A, B \in L^2(\lambda_\xi)$. The reflection forms are nonnegative on the entire positive-side $L^2(\lambda_\xi)$ spaces. Any specified measurable event of full parent measure has full measure in almost every component.

Additional Gaussian clause. If, for one $C \geq 0$, the parent law obeys

$$\mathbb{E}_\lambda e^{F(f)} \leq \exp\bigl(C\|f\|_2^2/2\bigr) \qquad (f \in \mathcal{S}(\mathbb{R}^3; V) \text{ real}),$$

the same common full-measure set can be chosen so that every component in it obeys this bound with the same C, simultaneously for all real Schwartz tests.

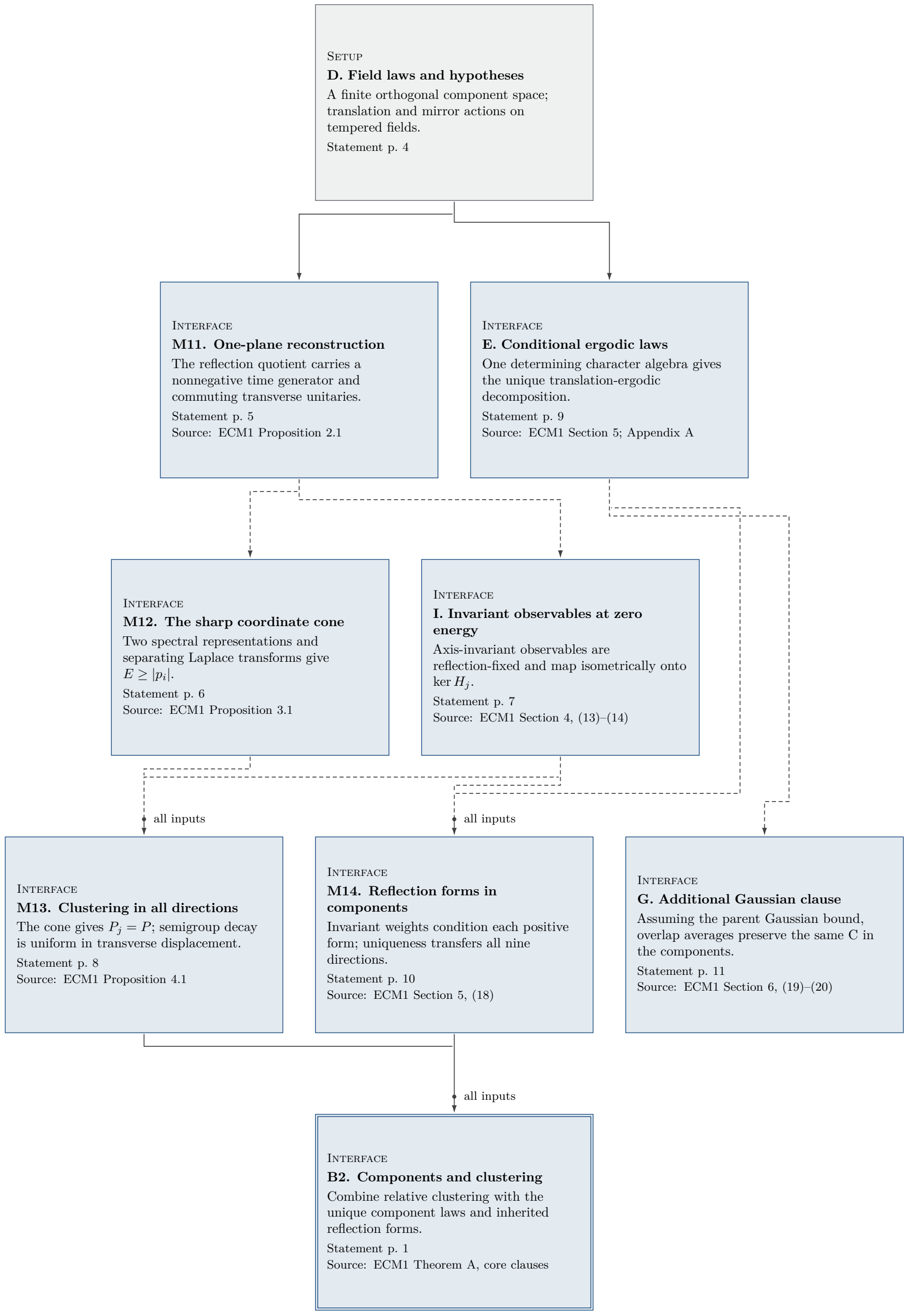
Proof organization. The spectral argument M11–M13 proves relative clustering. The conditional laws of E inherit the reflection forms by I and M14; applying M13 to these ergodic laws gives their factorized clustering. The overlap estimate in G supplies the additional Gaussian clause independently of the spectral cone. Intersect its full-measure set with the one from M14. The conclusion gives qualitative clustering, without a decay rate or a positive spectral gap.

Source. ECM1, Theorem A, pp. 1–2. ECM1 extracts the component and reflection arguments of the comprehensive QED3 manuscript, Section 7.

ECO1 – Continuum components and clustering

Definitions/setup | Model-specific | Interface (general)

ECO1 expands QEO1 B2 and M11–M14. Each box gives an ECM1 statement or a step in its supplied proof. The target box describes the core theorem; G gives its additional Gaussian clause.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.

Click a box for its statement. Return links are navigation, not proof dependencies.

D supplies notation and the hypotheses stated separately on each page. M14 carries the unique ergodic decomposition from E. B2 applies M13 both to the parent and to the component laws from M14. G assumes the parent Gaussian bound: it is a separate conclusion, not an input to the core B2 junction. Its full-measure set is intersected with the component set in B2.

2. D. Field laws and the application hypotheses

Setup. Overview Target.

Standing data. The field space, action, positive sides, and reflection form are those in B2. For an axis reflection r_j, write $\mathcal{R}_j = \mathcal{R}_{r_j}$, $\Theta_j A = \overline{\mathcal{R}_j A}$, and $\mathcal{E}_j^+ = L^2(\Sigma_{\{x_j>0\}}, \lambda)$. The inner product is conjugate linear in its first argument. Use P_j for the projection onto observables invariant under $U(te_j)$, and P for the projection onto all translation-invariant observables. The spectral momenta below are written P_i to distinguish them from these conditional expectations.

Hypotheses by proof task. Translation invariance is the assumption for E. A single invariant reflection-positive plane suffices for M11 and I. The cone in M12 uses that plane and one diagonal plane. The nine-plane hypotheses of B2 supply all the cones used in M13, and all the component reflections in M14. G assumes translation invariance and the displayed Gaussian bound with one constant C. These are conditions on an already given probability law.

Application to QED3. For the field-strength application, take $V = \Lambda^2\mathbb{R}^3$ with its induced orthogonal action. CEM1 supplies continuum laws and the inherited translation, reflection, and Gaussian properties; its hypotheses are checked in the QED3 assembly. The theorem B2 then supplies component clustering and the inherited bounds. Closedness passes as a specified measurable full-measure event. The further rotation and non-Gaussian cumulant conclusions are separate arguments in QEM1 and VTM1.

The representation ρ specifies how geometric motions act on the fields. The law hypotheses in B2 require the listed reflections and translations; invariance under every rotation is not an additional assumption here.

Source. ECM1, Introduction, especially (1)–(5), and the last paragraph of Section 6. Application: QEM1, Proposition 3.1 and its proof; VTM1, Theorem A, component hypotheses.

3. M11. Reconstruction at one reflection plane

Interface. Overview Target.

Statement. Suppose λ is translation invariant and invariant and reflection positive at $x_j = 0$. The form

$$b_j(A,B) = \mathbb{E}_\lambda[(\Theta_j A)B], \qquad A, B \in \mathcal{E}_j^+,$$

is positive semidefinite and $b_j(A,A) \le \|A\|_2^2$. Let $\mathcal{H}_j$ be its quotient completion and $[A]_j$ the quotient class. There are a nonnegative self-adjoint operator H_j and a strongly continuous unitary representation V_j of $e_j^\perp$, commuting with H_j, such that

$$e^{-tH_j} V_j(y)[A]_j = [U(te_j + y)A]_j \quad (t \ge 0,\ y \perp e_j), \qquad H_j[1]_j = 0.$$

Write $V_j(se_i) = e^{is\mathsf{P}_i}$ for $i \ne j$. Their joint spectral measure is denoted by E_j.

Proof tasks. Character sums are dense in the positive-side L^2 space: finite-dimensional Fourier uniqueness first gives density on finitely many evaluations, and increasing evaluation sigma-algebras give the full assertion. Thus the bounded form extends the assumed positivity. Continuity of translated tests and bounded convergence give strong continuity of translations, first on characters and then by density.

The quotient contraction needs a separate estimate. With $|A|_b = b_j(A,A)^{1/2}$, reflection and translation give

$$|U(te_j)A|_b \le |A|_b^{1-2^{-k}} \|A\|_2^{2^{-k}} \qquad (k \ge 1).$$

Let $k \to \infty$. This proves that positive translations descend through the null space as symmetric contractions. Their semigroup is positive since $T(t) = T(t/2)^2$; strong continuity makes it injective. Spectral calculus gives $T(t) = e^{-tH_j}$. Transverse translations preserve the form and have inverses, hence descend to the commuting unitaries.

Source. ECM1, Proposition 2.1, pp. 2–3; character density in Section 5, p. 6. Corresponds to QEO1 M11.

4. M12. The sharp coordinate cone from two mirrors

Interface. Overview Target.

Statement. Under the hypotheses of M11, fix $i \neq j$ and assume invariance and reflection positivity also at $x_j + x_i = 0$. Then

$$\mathsf{E}_j\{(E,p) : E < |p_i|\} = 0, \qquad H_j \geq |\mathsf{P}_i|$$

in the joint spectral calculus.

Two representations of one correlation. Fix $D > 0$ and $A \in L^2(\Sigma_B, \lambda)$, where $B = \{0 < x_j < D,\ |x_i| < D\}$. Put $\psi = [A]_j$ and let ϱ be its spectral measure projected to (E, p_i). With $z = E - ip_i$ and $a, c \geq 0$, coordinate reconstruction gives

$$K(a,c) = \mathbb{E}_\lambda[(\Theta_j A)U(2De_j + a(e_j + e_i) + c(e_j - e_i))A] = \int e^{-2DE} e^{-az - c\bar{z}}\, d\varrho.$$

Reconstruct the same correlation using the diagonal as time:

$$K(a,c) = \int e^{-aE' + icp'}\, d\varrho'(E', p'), \qquad E' \geq 0, \quad \|\varrho'\|_{\mathrm{TV}} \leq \|A\|_2^2.$$

Here a is the time parameter and c is transverse. Apply M11 to the transported law in rotated coordinates; equality of that law with λ is unnecessary. The supports and the factors $\sqrt{2}$ are checked in ECM1 (9)–(10).

Estimate and spectral support. For $h \in L^1(0,\infty)$, let $\widetilde{h}(z) = \int_0^\infty h(a) e^{-az}\, da$. Comparing the two representations yields

$$\int e^{-2DE} |\widetilde{h}(E - ip_i)|^2\, d\varrho \leq \|A\|_2^2 \sup_{u \geq 0} |\widetilde{h}(u)| \sup_{v \in \mathbb{R}} |\widetilde{h}(iv)|.$$

The supplied choice, for $s = \pm 1$ and integers $n \geq 1$, is

$$\widetilde{h}_n(z) = e^{-n} z^{1 - 4isn/\pi} (1+z)^{-3}, \qquad \widetilde{h}_n(0) = 0,$$

using the principal complex power. The two suprema are at most e^{-n} and e^n. On $s \arg z \geq \pi/4 + \delta$, its squared modulus is at least $e^{8\delta n/\pi} |z|^2 |1+z|^{-6}$. Letting $n \to \infty$ excludes each such sector from the positive measure ϱ. Both signs and rational $\delta > 0$ give the cone. The inverse-transform check uses the binomial expansion of $(1 - (1+z)^{-1})^{1 - 4isn/\pi}$, whose coefficients are absolutely summable, convolved with an integrable gamma kernel. Retain this check at the imaginary boundary. Density of local positive-side characters then makes the excluded spectral projection zero.

Source. ECM1, Proposition 3.1, pp. 3–4, especially (9)–(12). Corresponds to QEO1 M12.

5. I. Invariant observables and zero energy

Interface. Overview Target.

Statement. Assume only the one-plane hypotheses of M11. Every observable in $\operatorname{ran} P_j$ has a representative in either half-space bounded by any plane $x_j = c$, and is fixed by $\mathcal{R}_j$. The quotient map restricts to an isometry

$$\operatorname{ran} P_j \longrightarrow \ker H_j.$$

This map is onto. In particular, if $P_0 = \mathbf{1}_{\{0\}}(H_j)$, then

$$P_0[B]_j = [P_j B]_j \qquad (B \in \mathcal{E}_j^+).$$

Proof tasks. Approximate an axis-invariant observable by local character sums and translate each approximant into the chosen half-space. The approximation error is preserved because the observable itself is fixed. The half-space L^2 subspace is closed.

On these observables $\mathcal{R}_j$ is a positive self-adjoint involution: positivity is precisely the reflection inequality in the half-space representative. Its spectrum is therefore $\{1\}$. The reflection form becomes the ordinary L^2 inner product, which proves the isometry.

Average the identity

$$\langle [A]_j, e^{-tH_j}[B]_j \rangle = \langle \mathcal{R}_j A, U(te_j)B \rangle$$

over $0 \le t \le T$. The spectral theorem gives P_0 on the left; the translation mean-ergodic limit gives P_j on the right. This proves the displayed projection identity. Its range is dense in $\ker H_j$ and is closed by the isometry, proving surjectivity.

The spectral cone enters later, in M13. Here the invariant weight argument needed by M14 already follows from one plane.

Source. ECM1, Section 4, p. 5, through (14); translation averages in Lemma A.2, p. 8. This isolates the first step of QEO1 M13.

6. M13. Clustering uniformly in transverse displacement

Interface. Overview Target.

Statement. Under the nine-plane hypotheses of B2, $P_1 = P_2 = P_3 = P$. For every $A, B \in L^2(\lambda)$ and every j,

$$\lim_{|t|\to\infty} \sup_{y \perp e_j} |\langle A, U(te_j + y)(1 - P)B\rangle| = 0.$$

Consequently $\langle A, U(a)B\rangle \to \langle A, PB\rangle$ as $|a| \to \infty$. If λ is translation ergodic, the limit factorizes and $\ker H_j = \mathbb{C}[1]_j$.

Zero energy forces full translation invariance. The cones in M12 put all transverse momenta at zero on $\ker H_j$. For $A \in \operatorname{ran} P_j$ and $y \perp e_j$, $[U(y)A - A]_j = 0$. This difference remains in $\operatorname{ran} P_j$; the isometry in I implies $U(y)A = A$. Hence each $P_j = P$.

The uniform bound. For $A = \mathcal{R}_j A'$ with $A', B \in \mathcal{E}_j^+$, the reconstruction and the projection identity of I give

$$|\langle A, U(te_j + y)(1 - P_j)B\rangle| \le \|[A']_j\| \, \|e^{-tH_j}(1 - P_0)[B]_j\|.$$

Spectral dominated convergence makes the right side tend to zero. It is independent of y. Approximate general observables by local character sums and move their supports to opposite half-spaces; this changes t by a fixed amount. The L^2 approximation error is uniform in every displacement. Adjoints give negative t.

Along any sequence $|a| \to \infty$, pass to a subsequence with one fixed coordinate tending to infinity in modulus. The uniform transverse bound applies along that subsequence. Repeating this for subsequences proves the full limit. For an ergodic law the projection P is onto constants; I then also identifies the one-dimensional zero-energy space.

Source. ECM1, (15) and Proposition 4.1, p. 5. Corresponds to the clustering part of QEO1 M13.

7. E. Conditional laws and the common ergodic decomposition

Interface. Overview Target.

Statement. For every translation-invariant law on $\mathcal{X}$, regular conditional laws λ_F given $\mathcal{I}$ exist, with

$$\lambda_F(\Phi) = \mathbb{E}_\lambda[\Phi \mid \mathcal{I}](F), \qquad \lambda = \int \lambda_F \, d\lambda(F).$$

The first identity holds almost surely for each integrable Φ, and for each nonnegative Φ by monotone convergence. The distribution of λ_F is the unique decomposition into translation-invariant ergodic laws. A single full-measure set supports these properties. Every specified measurable event of full parent measure has full measure in almost every component.

Measurability and a determining family. ECM1, Lemma A.1, realizes $\mathcal{X}$ by countably many test evaluations as a countable union of closed subsets of a countable product of lines. It is standard Borel, and translations are jointly measurable. Choose the countable rational character algebra of ECM1, Section 5, stable under rational translations and cubic isometries. Fourier uniqueness makes it determining and dense in L^2 for every law on the evaluation sigma-algebra.

Ergodicity and uniqueness. Cube averages $M_T = T^{-3} \int_{[0,T]^3} U(a)\, da$ converge strongly to P. Choose times T_k with summable squared errors simultaneously for the first k members of the countable algebra. Conditioning these errors gives convergence in $L^2(\lambda_F)$ on one full set. The variable PA has zero conditional variance and equals the constant $\lambda_F(A)$ there. Testing translation identities against invariant weights first proves invariance for rational translations; continuity extends it to all translations. The mean-ergodic theorem inside each conditional law now makes its invariant projection the constant projection.

For any other ergodic decomposition, the same summable parent errors give an almost-sure limit, while the component averages converge to their expectations in L^2. Uniqueness of limits in probability identifies the expectations on the determining algebra, hence the component law itself. This proves uniqueness of the distribution of conditional laws. The event assertion is the conditional identity applied to its indicator.

Source. ECM1, Section 5, p. 6, (17), and Appendix A, pp. 7–8. The conditional-distribution input is cited there to Durrett, *Probability: Theory and Examples*, fifth edition, Theorems 4.1.16–4.1.17. This expands the conditional-law part of QEO1 M14.

8. M14. Reflection forms in the same ergodic components

Interface. Overview Target.

Statement. Under the hypotheses of B2, the unique ergodic decomposition in E has one full-measure set on which every component is invariant and reflection positive at all real translates of the nine planes. Positivity holds on the full positive-side L^2 spaces. The components are cubically invariant.
Conditioning a positive form. Fix an axis mirror. For bounded, nonnegative, $\mathcal{I}$-measurable h, I gives a half-space representative and $\mathcal{R}_j h = h$. Apply the parent form to $\sqrt{h}A$, obtaining

$$\mathbb{E}_\lambda[h(\Theta_j A)A] \geq 0.$$

Reflection invariance similarly gives $\mathbb{E}_\lambda[h\mathcal{R}_j A] = \mathbb{E}_\lambda[hA]$ for bounded invariant weights. All such weights are admissible; these inequalities and identities are therefore conditional ones. Impose them first on the countable determining character families from E. In each conditional law, density and boundedness of the form extend positivity to the full half-space L^2 space. Translation invariance moves the plane to every real height.
All nine directions. For another mirror, transport the parent law by an orthogonal map taking its normal to an axis. Apply the preceding one-plane argument to that transported law, and transport its component laws back. Uniqueness in E identifies their distribution with the original ergodic decomposition. Intersect the nine full-measure sets. The nine reflections generate the cubic group, which proves the asserted invariance of each component.

At this point every component on that set satisfies the hypotheses of M13. Apply that theorem within each component to obtain factorized clustering for all its L^2 observables. This last application is part of the assembly in B2; the conditional reflection argument itself uses only I and E.

Source. ECM1, Section 5, pp. 6–7, especially (18). This is the reflection-inheritance part of QEO1 M14.

9. G. The same Gaussian constant after conditioning

Interface. Overview Target.

Statement and additional hypothesis. Let λ be translation invariant and satisfy the Gaussian bound in B2 with $C \geq 0$. Its conditional laws from E satisfy that bound with the same C, simultaneously for all real Schwartz tests, on one full-measure set. Reflection positivity is not needed in this step.
Spatial averages of an exponential. Fix real f, put $Z = e^{F(f)}$, $c = e^{C\|f\|_2^2/2}$, and $\phi = PZ$. For every integer $n \geq 2$, the parent bound gives

$$\mathbb{E}_\lambda[(M_T Z)^n] \leq T^{-3n} \int_{([0,T]^3)^n} \exp\left(\frac{Cn}{2}\|f\|_2^2 + \frac{C}{2}\sum_{p\neq q}\langle f, \tau_{a_p - a_q} f\rangle\right) da.$$

The overlaps tend to zero at large separation and are bounded by $\|f\|_2^2$. For fixed n, exclude the configurations with one pair of points within distance δ. Their relative volume is at most $\binom{n}{2}(2\delta)^3 T^{-3}$, while the integrand is bounded by $e^{Cn^2\|f\|_2^2/2}$. On the complement it is at most $c^n e^{Cn(n-1)\epsilon/2}$ once δ is large enough.
From moments to a conditional bound. All moments of Z are finite. Cube averages converge to ϕ in L^2; their uniform L^{2n} bounds upgrade this to L^n convergence. First let $T \to \infty$, then $\epsilon \to 0$, to obtain $\|\phi\|_n \leq c$ for every n. Let $n \to \infty$, giving $\phi \leq c$ almost surely. The conditional identity in E yields the desired bound for this test.

Intersect the full-measure sets for a countable real Schwartz-dense test family. For each remaining test, continuity of every sampled tempered distribution and Fatou's lemma extend the inequality. When the parent also satisfies B2's reflection hypotheses, intersect this set with the set from M14. All the stated properties then hold in the same components.

Source. ECM1, Section 6, p. 7, (19)–(20). This is the Gaussian-inheritance part of QEO1 M14 and the component bound used in VTM1.